# Area - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

## Question 1

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 10*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | eg $\pi\times\left(\frac{14}{2}\right)^2$ oe or $\pi\times7^2$ oe or $49\pi$ | | 2 | M1 |
| | | 154 | | A1 accept 153.86 – 154 |
| | | | | **Total 2 marks** |

## Question 2

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 13*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | eg $\pi\times6.4^2\left(=\frac{1024}{25}\pi\right)$ | | 2 | M1 allow 3.14… or $\frac{22}{7}$ for π |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 129 | | A1 accept 128 – 129 |
| | | | | **Total 2 marks** |

## Question 3

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2021 · Question 9*

| Question | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | (0.5 × 4 × 6) – (0.5 × 2 × 3)<br>or 2 × 3 + 0.5 × 2 × 3<br>or (0.5 × 4 × 6) –<br>(0.25 × "0.5 × 4 × 6") oe | | 3 | M2 if not M2 then M1 for either<br>0.5 × 4 × 6 (= 12) or 0.5 × 2 × 3 (= 3)<br>if not M2 then M1 for either area of 1 large parallelogram (2 × 3) or 1 triangle (0.5 × 3 × 2)<br>Allow M1 for consistent use of incorrect side lengths, eg 0.5 × 5 × 7 |
| | | 9 | | A1 |
| | | | | **Total 3 marks** |

## Question 4

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 15*

| | | | | | |
|---|---|---|---|---|---|
| **15** | $\frac{1}{2}\times8.9\times6.7\times\sin74°$ or 28.6(600...) | $h=6.7\sin74°$ or 6.44(0...) | | 3 | M1 or a complete correct method to find the perpendicular height |
| | × 2 | 8.9×"6.44(0...)" | | | M1 (dep) for a complete method to find area of parallelogram |
| | | | 57.3 | | A1 for answer rounding to 57.3 (57.320...) |
| | | | | | **Total 3 marks** |

# Area - Mark scheme

## Question 5

*Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 19*

| | | | | | |
|---|---|---|---|---|---|
| **19** | $0.5 \times 6.4 \times 9.7 \times \sin 110$ (= 29.16…) | | | M1 | M2 for $6.4 \times 9.7 \times \sin 110$ |
| | $2 \times$ "29.16…" | | | M1 | |
| | | 58.3 | 3 | A1 | for 58.3 – 58.4 |
| | **alternative** | | | | |
| | $AC = \sqrt{6.4^2 + 9.7^2 - 2\times 9.7\times 6.4\times\cos 110}$ (=13.323...)<br>$DAC = \sin^{-1}(\frac{\sin 110}{'13.323'}\times 9.7)$(= 43.167...) or<br>$ACD = \sin^{-1}(\frac{\sin 110}{'13.323'}\times 6.4)$(= 26.83...) | | | M1 | For method to find *AC* and angle *DAC* or angle *ACD* |
| | Area = (sin '43.167..' × 6.4 × 2 × '13.323..') ÷ 2<br>Or area = (sin '26.83..' × 9.7 × 2 × '13.323...') ÷ 2 | | | M1 | find *DB* and then area using half product of diagonals |
| | | 58.3 | | A1 | for 58.3 – 58.4 |
| | | | | | **Total 3 marks** |

## Question 6

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 13*

| Q | Working | Answer | Mark | Notes | |
|---|---|---|---|---|---|
| | eg $2.5 \times 6.5$ (= 16.25)<br>**or** $0.5 \times 6.5 \times 1$ (= 3.25)<br>**or** $3.5 \times 6.5$ (= 22.75) | | 4 | M1 | M2 for $0.5(2.5 + 3.5) \times 6.5$ (= 19.5)<br>**or**<br>$2 \times (0.5(2.5 + 3.5) \times 6.5)$<br>(= 39) |
| | $2.5 \times 6.5 + 0.5 \times 6.5 \times 1$ (= 19.5)<br>**or** $2 \times (2.5 \times 6.5 + 0.5 \times 6.5 \times 1)$ (= 39)<br>**or** $3.5 \times 6.5 - 0.5 \times 6.5 \times 1$ (= 19.5)<br>**or** $2 \times (3.5 \times 6.5 - 0.5 \times 6.5 \times 1)$ (= 39) | | | M1 | |
| | $2 \times$ "19.5" ÷ 12 (= 3.25)<br>**or** "39" ÷ 12 (= 3.25)<br>**or** 12 + 12 + 12 + 12 (= 48)<br>**or** $4 \times 12$ (= 48) | | | M1 | **or** [their area] ÷ 12 (dep on M1)<br>**or** using multiples of 12 for [their area]<br>eg area = 19.5 **and** 12 + 12 (= 24) **or** $2 \times 12$ (= 24) |
| | | 4 | | A1 | dep on M2, must be from correct working |
| | | | | | **Total 4 marks** |

## Question 7

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2018 · Question 1*

| | | | | | |
|---|---|---|---|---|---|
| **1** | $20 \times 14$ (= 280) | 460 | 4 | M1 | |
| | $\frac{20+16}{2} \times (24-14)$ (= 180) | | | M1 | |
| | "280" + "180" | | | M1 | (dep) on at least one of the previous M marks |
| | | | | A1 | |
| | | | | | **Total 4 marks** |
| | **Alternative scheme 1** | | | | |
| | (24 +14)÷2 (= 19) **and** (20−16)÷2 (=2) | 460 | 4 | M1 | |
| | $2 \times 19$ (= 38) **and** $16 \times 24$ (= 384) | | | M1 | |
| | "38" + "38" + "384" | | | M1 | (dep) on at least one of the previous M marks |
| | | | | A1 | |
| | | | | | **Total 4 marks** |
| | **Alternative scheme 2** | | | | |
| | $20 \times 24$ (= 480) | 460 | 4 | M1 | |
| | (20−16)÷2 (=2) **and** 24 − 14 (= 10)<br>$2 \times 10 = 20$ | | | M1 | |
| | "480" – "20" | | | M1 | (dep) on at least one of the previous M marks |
| | | | | A1 | |
| | | | | | **Total 4 marks** |

# Area - Mark scheme

## Question 8

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 3*

| | | | | | |
|---|---|---|---|---|---|
| **3** | | E.g.<br>$12 \times 9$ (=108) **or** $(9-6) \times x$ (= $3x$) | | 4 | M1 for one correct relevant area |
| | | E.g.<br>$129 - $ '108' (= 21) **or**<br>'108' + '$3x$' = 129 | | | M1 (dep on M1) for 129 used correctly with another area<br>**or**<br>for a correct equation (ft) with bracket(s) expanded |
| | | E.g.<br>'21' ÷ $(9-6)$ **or**<br>$x = \frac{129-'108'}{9-6}$ | | | M1 for a complete method |
| | | | 7 | | A1 Accept 7 cm |
| | | | | | **Total 4 marks** |

## Question 9

*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 4*

| | | | | |
|---|---|---|---|---|
| **4.** | eg $((15\times12)+(\frac{1}{2}\times12\times10)-(\frac{1}{2}\times12\times4)$ or<br>$180+60-24$ or<br>$(10+15)\times12-(\frac{1}{2}\times12\times4)+(\frac{1}{2}\times10\times6)+(\frac{1}{2}\times10\times6)$<br>or $300-(24+30+30)$ or<br>$2\times\frac{1}{2}(15+21)\times6$ or $2\times108$<br><br>eg $(\frac{1}{2}\times4\times12)$ and $(\frac{1}{2}\times10\times6)$ (24 and 30) or<br>$(\frac{1}{2}\times4\times12)$ and $(\frac{1}{2}\times10\times12)$ (24 and 60) or<br>$(\frac{1}{2}\times4\times6)$ and $(\frac{1}{2}\times10\times6)$ (12 and 30) or<br>$(\frac{1}{2}\times4\times6)$ and $(\frac{1}{2}\times10\times12)$ (12 and 60) or<br>$\frac{1}{2}(15+21)\times6$ or 108 or<br>$\frac{1}{2}(15+11)\times6$ or 78<br><br>eg $(\frac{1}{2}\times4\times6)$ or 12 or $(\frac{1}{2}\times4\times12)$ or 24 or<br>$(\frac{1}{2}\times10\times6)$ or 30 or $(\frac{1}{2}\times10\times12)$ or 60 or<br>$(\frac{1}{2}\times11\times6)$ or 33 or $(\frac{1}{2}\times11\times12)$ or 66 or<br>$(\frac{1}{2}\times15\times6)$ or 45 or<br>$15\times6$ or 90 or $15\times12$ or 180 or<br>$25\times6$ or 150 or $25\times12$ or 300 or<br>$10\times6$ or 60 or $10\times12$ or 120 or<br>$11\times6$ or 66 or $11\times12$ or 132 or<br>$4\times12$ or 48 or $4\times6$ or 24 | | 4 | M3 For a complete method.<br><br>If not M3 then M2 for 2 different but non overlapping triangles or 1 trapezium<br><br>If not M2 then M1 for a correct area of a triangle or rectangle.<br><br>**Note: The lists of examples are not exhaustive.** |
| | | 216 | | A1 |
| | | | | **Total 4 marks** |

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## Question 10

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 17*

| Question | | Working | Answer | Mark | Notes |
|---|---|---|---|---|---|
| | | $175 \div 25\ (=7)$ **or**<br>$25 \times h = 175$ | | 4 | M1 method to find height of the parallelogram |
| | | $\frac{11+25}{2} \times$ "7" $(=126)$ oe **or**<br>$\frac{11+25}{2} \times$ [their height of parallelogram] | | | M1 method to find the area of the trapezium ft their value for the height of the parallelogram if clearly identified on the diagram or in working space |
| | | $\frac{"126"}{175}\ (=0.72)$ **or**<br>$\frac{"126"}{175} \times 100$ **or**<br>$\frac{\text{[their area of trapezium]}}{175}\ (\times 100)$ | | | M1 dep on M1 method to find the area of the trapezium as a fraction **or** percentage of the area of the parallelogram ft their value for the area of the trapezium as long as clearly identified as an area on the diagram or in the working |
| | | *Correct answer scores full marks (unless from obvious incorrect working)* | 72 | | A1 |
| | | | | | **Total 4 marks** |

## Question 11

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 5*

| | | | | | |
|---|---|---|---|---|---|
| 5 | $28 \times 12\ (=336)$ or $5 \times 12\ (=60)$ or $18 \times 12\ (=216)$<br>or<br>$28 \times 20\ (=560)$ or $\frac{1}{2}(CD+"18")"8"$ oe eg $72+4CD$<br>[numbers in " " come from correct working]<br><br>**Check diagram for areas** | | 4 | M1 | For a correct method to find the area of a rectangle (may be seen as part calculation) or a correct expression for the area of the trapezium with numbers substituted.<br><br>Allow for other correct methods to find area linked to this shape. |
| | "336" + 0.5("18"+ $CD$)"8" = 434 oe eg<br>4("18" + $CD$) = 98<br>or<br>eg 0.5("18" + $CD$)"8" = "98" oe eg $\frac{1}{2}(18+CD)=12.25$<br>or<br>$"560"-2\left(0.5(5+x)"8"\right)=434$ oe (where $x$ is horizontal from $D$ to perp with $AF$)<br><br>[numbers in " " come from correct working] | | | M1 | correct use of their values from correct working for an equation involving $CD$ ($CD$ could be labelled with any letter) |
| | eg $(CD=)\frac{196-144}{8}\left(=\frac{52}{8}\right)$ or $(CD=)\frac{98-72}{4}\left(=\frac{26}{4}\right)$<br>or $(CD=)\frac{434+152-560}{4}$ or $(CD=)2\times12.25-18$ or<br>$98\times2(=196)$, $"196"\div8(=24.5)$, $"24.5"-18$ | | | M1 | a correct process to solve a correct equation **or** a correct process to find $CD$ using **correct values** |
| | | 6.5 | | A1 | oe |
| | | | | | **Total 4 marks** |

## Question 12

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2021 · Question 18*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | $8 \times x$ (= $8x$) **or** $14 \times x$ (= $14x$) **or** $(14-8) \times x$ (= $6x$) **or** $\frac{1}{2} \times (14-8) \times (13-x)$ (= $39-3x$) **or** $\frac{13+x}{2} \times (14-8) (= 39+3x)$ **or** $\frac{1}{2} \times 13 \times (14-8)$ (=39) **or** $\frac{8+14}{2} \times x$ (= $11x$) **or** $14 \times 13$ (=182) **or** $8 \times (13-x)$ (= $104-8x$) **or** $\left(\frac{8+14}{2} \times (13-x)\right)$ (= $143-11x$) oe | | 4 | M1 one correct area linked to the shape |
| | $14x + 6 \times \frac{1}{2} \times (13-x)$ oe eg $8x + \frac{x+13}{2} \times 6$ **or** $\frac{8+14}{2} \times x + \frac{13 \times (14-8)}{2}$ **or** "182" $- \left(\frac{8+14}{2} \times (13-x)\right)$ **or** $11x + 39$ oe | | | M1 ft from correct working expression for total area of shape – with no parts omitted or duplicated<br><br>Adding up parts of given shape **or**<br>Large rectangle subtracting trapezium (or subtracting (rectangle + triangle)) |
| | eg $11x + 39 = 91.8$ **or** $14x + 39 - 3x = 91.8$ **or** "182" $- 143 + 11x = 91.8$ or $16x + 6x + 78 = 183.6$ oe | | | M1 fully correct equation with no fractions (allow 91.8 or multiples of 91.8 but no other decimals) **and** no brackets |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | 4.8 | | A1 or $4\frac{4}{5}$ oe or $\frac{24}{5}$ oe |
| | | | | **Total 4 marks** |

## Question 13

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 7*

| | | | | | |
|---|---|---|---|---|---|
| 7 | eg $13 \times 21$ (=273) **or** $21 \times h$ (= $21h$) **or** $0.5(15+21) \times y$ **or** $15(h-13)$ **or** $2 \times \frac{1}{2}(3(h-13))$ **or** $\frac{1}{2}(13+h) \times 3$ (= $19.5 + 1.5h$) **or** $15 \times h$ (= $15h$) | | 4 | M1 | A correct calculation for an area linked to the shape. ($h-13$) might be written as $x$ or $y$ etc: this is acceptable (even allow $h$) [allow without brackets for this mark only] |
| | eg $390 -$ "273" (= 117) **or** $13 \times 21$ **and** $0.5(15+21)(h-13)$ **or** $13 \times 21$ **and** $0.5(15+21)y$ oe **or** $21h$ **and** $2 \times \frac{1}{2}(3(h-13))$ oe **or** $13 \times 21$ **and** $15(h-13)$ **and** $2 \times \frac{1}{2}(3(h-13))$ oe **or** $2 \times \frac{1}{2}(13+h) \times 3$ **and** $15 \times h$ | | | M1 | For considering the area of all parts of the shape (parts need not be added or subtracted for the whole shape)<br><br>(where $y$ = height of $BCDE$ )<br><br>($h-13$) might be written as $x$ or $y$ etc: this is acceptable (even allow $h$)<br><br>[correct use of brackets] |
| | "117" $\div (0.5 \times (15+21))$(= 6.5) or **or** $\frac{1}{2}(15+21) \times y =$ "117" **or** $273 + 18(h-13) = 390$ **or** $15(h-13) + 2 \times \frac{1}{2}(3(h-13)) =$"117" oe **or** $2 \times \frac{1}{2}(13+h) \times 3 + 15h = 390$ **Typical equations here simplify to :** $18y = 117$, $18h - 234 = 117$, $18h + 39 = 390$, $18h = 351$ | | | M1 | A correct calculation to find height of trapezium or height of shape or a correct equation involving height of trapezium or height of shape or 6.5 ($h-13$) might be written as $x$ or $y$ etc: this is acceptable (even allow $h$)<br><br>[correct use of brackets] |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 19.5 | | A1oe | eg $\frac{39}{2}$ |
| | | | | | **Total 4 marks** |

## Question 14

*Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 7*

| | | | | | |
|---|---|---|---|---|---|
| 7. | Splits shape appropriately<br>eg rectangle + triangle or rectangle + trapezium or 'completing the rectangle' | | 4 | B1 | If lines not present on diagram then can be implied by correct method for at least two areas (areas must not overlap and must not be contradictory) |
| | eg. $9 \times 10$ or 90 or $9 \times 4$ or 36<br>or $9 \times 6$ or 54 or $\frac{1}{2} \times 7 \times 6$ or 21<br>or $\frac{1}{2} \times (16+9) \times 6$ or 75<br>$16 \times 10$ or 160 or $\frac{1}{2} \times (4+10) \times 7$ or 49 | | | M1 | for area of one appropriate rectangle, triangle or trapezium |
| | eg. $\frac{1}{2} \times 7 \times 6 + 9 \times 10$<br>$\frac{1}{2} \times 7 \times 6 + 9 \times 4 + 9 \times 6$<br>$9 \times 4 + \frac{1}{2} \times (16+9) \times 6$<br>$16 \times 10 - \frac{1}{2} \times (4+10) \times 7$ | | | M1 | for complete method |
| | | 111 | | A1 | cao |
| | | | | | **Total 4 marks** |

## Question 15

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 8*

| | | | | | |
|---|---|---|---|---|---|
| 8 | $8.5^2 - (8 \div 2)^2$ (= 56.25) **or** $\cos x = \frac{4}{8.5}$ oe | | | M1 | **or** eg $\cos A = \frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}$ |
| | $\sqrt{\text{"56.25"}}$ (= 7.5) **or** $x = \cos^{-1}\left(\frac{4}{8.5}\right)$ (= 61.927...)<br>oe | | | M1 | **or** eg $(A =) \cos^{-1}\left(\frac{8^2 + 8.5^2 - 8.5^2}{2 \times 8 \times 8.5}\right)$ (61.927...)<br>(other angle = 56.144...) |
| | $8 \times$ "7.5" $\div 2$ oe **or** $0.5 \times 8 \times 8.5 \times \sin$ "61.927..." | | | M1 | **or** eg $0.5 \times 8.5 \times 8 \times \sin$"61.927..." oe |
| | | 30 | 4 | A1 | |
| | | | | | **Total 4 marks** |

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## Question 16

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 24*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | $17.5^2 - 14^2$ (= 110.25) | | 4 | M1 or for use of cosine rule to find one of the angles<br>eg $28^2 = 17.5^2 + 17.5^2 - 2 \times 17.5 \times 17.5 \times \cos A$<br>or eg $\cos B = \frac{14}{17.5}$ |
| | $\sqrt{17.5^2 - 14^2}$ (=10.5) | | | M1 or for rearranging the cosine rule to<br>eg $\cos A = \frac{17.5^2 + 17.5^2 - 28^2}{2 \times 17.5 \times 17.5}$ ($A$ = 106.26...)<br>or eg $B = \cos^{-1}(\frac{14}{17.5})$ (= 36.86...) |
| | 0.5 × 28 × "10.5" oe | | | M1 or for 0.5 × 17.5 × 17.5 × sin106.26... oe<br>eg 0.5 × 17.5 × 28 × sin36.86...<br><br>[clear use of Heron's formula:<br>M1 for S = 0.5(17.5 + 17.5 + 28)(=31.5)<br>M2 for $\sqrt{"31.5"("31.5"-17.5)^2("31.5"-28)}$ oe] |
| | | 147 | | A1 accept awrt 147 |
| | | | | **Total 4 marks** |

## Question 17

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2021 · Question 16*

| Q | Working | Answer | Mark | Notes |
|---|---|---|---|---|
| | 2 × 0.75 (= 1.5) oe or 2 × 0.75 × 2 (= 3) oe | | 5 | M1 for area of rectangle |
| | $\pi \times (0.5 \div 2)^2$ (= 0.1963) or $\frac{1}{2} \times \pi \times (0.5 \div 2)^2$ (= 0.09817) | | | M1 for area of circle or area of semicircle |
| | "1.5" – "0.09817" (= 1.4018…) or "3" – "0.1963" (= 2.8036…) | | | M1 |
| | "1.4018" × 2 × 250 ÷ 4 (= 175.228…) or "2.8036" × 250 ÷ 4 (= 175.228…) or "1.4018" × 250 ÷ 4 (= 87.6…) | | | M1or for 87 – 88 |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | 175 | | A1 Allow 175 – 176 |
| | | | | **Total 5 marks** |

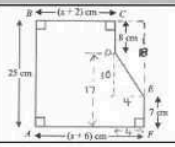
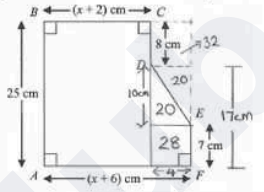
## Question 18

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2019 · Question 9*

| | | | | | |
|---|---|---|---|---|---|
| **9** | | 0.5 × 6 × 6 (=18) | | 5 | M1 For area of triangle, or may use $\frac{1}{2} \times 6 \times 6\sqrt{2} \sin 45$ or $\frac{1}{2} \times 6\sqrt{2} \times 3\sqrt{2}$ oe |
| | | $(d^2 =) 6^2 + 6^2$ (=72) or $\frac{AC}{(\sin 90)} = \frac{6}{\sin 45}$ | | | M1 |
| | | $\sqrt{6^2 + 6^2}$ ($= \sqrt{72} = 6\sqrt{2}$ =8.4(85...)or 8.5) **or** $AC = \frac{6(\sin 90)}{\sin 45} = 6\sqrt{2}$ =8.4(85...)or 8.5) oe | | | M1 |
| | | $0.5 \times \pi \times \left(\frac{"8.48.."}{2}\right)^2$ (= 9π or 28....) | | | M1 |
| | | | 46.3 | | A1 for 46.2 – 46.3 |
| | | | | | **Total 5 marks** |

## Question 19

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 10*

| Q | Working | An | Mark | Notes |
|---|---|---|---|---|
| 10 |  4 from F to left, vertically below D<br>17 from D vertically down to left of F<br>10 from D vertically to left of E<br>18 vertically up from E to right of C | | 5 | B1 for a length of 4 (cm) or 17 (cm) or 25 – 7 – 8 (=10 (cm)) or 18 (cm) in the correct place on the diagram or calculated or used correctly in working. |
| | eg $25(x+2)(=25x+50)$ **or** $\frac{7+(25-8)}{2}\times((x+6)-(x+2))(=48)$ **or**<br>$25(x+6)(=25x+150)$ **or** $\left(\frac{8+(25-7)}{2}\times 4\right)(=52)$ **or**<br>$8(x+2)(=8x+16)$ **or** $\frac{(x+2)+(x+6)}{2}\times(25-8-7)(=10x+40)$ **or**<br>$7(x+2)(=7x+14)$ **or** $10(x+2)(=10x+20)$ or<br>$7(x+6)(=7x+42)$ **or** $(25-7)\times(x+2)(=18x+36)$ **or** $0.5(25-8-7)\times 4(=20)$ | | | M2 for 2 correct expressions or values for the area of any 2 parts of the shape that do not overlap (unless subtracting) (need not be added or subtracted) (figures to be correct or come from correct working to award marks)<br>**(M1 for one correct expression)**<br>  |
| | eg $25x + 98 = 258$ or<br>$25(x + 2) + 4 \times 7 + 0.5 \times 4 \times 10 = 258$ oe or<br>$25(x + 6) = 258 + 20 + 32$ oe<br>**or** a fully correct numerical method eg $(258 - 98) \div 25$ oe | | | M1 for an equation that is correct or from correct working. This need not have expanded terms and may not equal 258 if other work has been done. All parts for their method must be included with no overlaps OR a complete numerical method |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | 6.4 | | A1 oe eg $160/25$ |
| | | | | **Total 5 marks** |

## Question 20

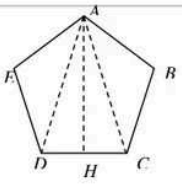
*Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 12*

| | | | | |
|---|---|---|---|---|
| 12 | $\sqrt{9.5^2-7.6^2}$ **or** $\sqrt{90.25-57.76}$ **or** $\sqrt{32.49}$ **or** $\sqrt{32.5}$ | | 5 | M1 |
| | $(BC=)$ 5.7 | | | A1 |
| | $\frac{1}{2}\times 7.6\times$"5.7" **or** 21.6(6) **or** 21.7 | | | M1 dep on first M1<br>or eg. $ACB=\sin^{-1}\left(\frac{7.6}{9.5}\right)(=53.1...)$<br>**and**<br>$\frac{1}{2}\times 9.5\times$"5.7"$\times\sin$"53.1" |
| | $\frac{1}{2}\times\pi\times\left(\frac{"5.7"}{2}\right)^2$ **or** 12.7(587...) **or** 12.8 | | | M1 dep on first M1 |
| | | 34.4 | | A1 for answer rounding to 34.4<br>($\pi\rightarrow$ 34.4187... 3.14→34.4123...) |
| | | | | **Total 5 marks** |

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## CHALLENGE QUESTIONS

### Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 22*

| | | | | | |
|---|---|---|---|---|---|
| 22 | $12^2 + 8^2 - 2\times12\times8\times\cos(105)$ (=257...) | | | M1 | |
| | 257(....) or $\sqrt{257}$ (=16.05..) | | | A1 | for 257 or awrt 258 or 16 - 16.1<br>If M1 has been awarded then allow the use of the candidate's value for *AD* in all subsequent working |
| | eg.<br>  <br><br>$(AH =)\sqrt{"16.05.."^2 - 6.5^2}$ (=14.6...) **or**<br>$(ADC =)\cos^{-1}\left(\frac{"16.05"^2 + 13^2 - "16.05"^2}{2\times"16.05"\times13}\right)$ (=66.08..) | | | M1 | (dep on first M1)<br>complete method to find height of pentagon or any angle within triangle *ADC*<br>E.g.<br>angle *ADC* = angle *ACD* = 66.08…<br>angle *DAC* = 47.8…<br>angle *DAH* = angle *CAH* = 23.9…<br>(accept all these angles rounded or truncated to 3 or more sig figs) |
| | eg.<br>0.5×12×8×sin(105) (=46.3...) **or**<br>12×8×sin(105) (=92.7…) **or**<br>0.5×13×"14.6" (=95.4...) **or**<br>0.5 × 13 × "16.05" × sin ("66.1") | | | M1 | any one relevant area<br>(any calculated values used must come from a correct method) |
| | eg.<br>2 ×0.5×12×8×sin(105) + 0.5×13×"14.6" **or**<br>2×0.5×12×8×sin(105) + 0.5 × 13 × "16.05" × sin ("66.1") | | | M1 | (dep on first M1)<br>complete correct method |
| | | 188 | 6 | A1 | accept answer in range 188 – 188.5 |
| | | | | | **Total 6 marks** |

### Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 9*

| | | | | | |
|---|---|---|---|---|---|
| 9 | eg $12\sin60\left(=6\sqrt{3}=10.3(9...)\right)$ **or** $\sqrt{12^2 - "6"^2}\left(=6\sqrt{3}=10.3(9...)\right)$<br>**or** (Area *ADC*) $=\frac{1}{2}\times12\times47\times\sin60\,(=244.2...)$ | | 5 | M1 | for a method find the height of the trapezium<br>**or** the area of triangle *ADC*<br><br>The first two M1 marks can be awarded in either order |
| | eg $12\cos60\left(=6\right)$ **or** $\sqrt{12^2 - \left("6\sqrt{3}"\right)^2}\left(=6\right)$ | | | M1 | (indep)<br>for a method find the base of the triangle, condone missing brackets around $"6\sqrt{3}"$<br><br>The first two M1 marks can be awarded in either order |
| | eg (*AB* =) 47 – "6" – "6" (= 35) | | | M1 | (dep on previous M1)<br>for method to find the length of *AB* |
| | eg (Trapezium =) $\frac{1}{2}\times(47+"35")\times"10.3(9...)"$<br>**or** (Rectangle + 2 × Triangle =) $"35"\times"10.3(9...)" + 2\times\frac{1}{2}\times"6"\times"10.3(9...)"$<br>**or** (Rectangle + 2 × Triangle =) $"35"\times"10.3(9...)" + 2\times\frac{1}{2}\times"6"\times12\times\sin60$<br>**or** (Triangle *ADC* + Triangle *ABC* =) $"244.2..."+\frac{1}{2}\times12\times"35"\times\sin120$<br>oe eg $(47-"6")\times"10.3(9...)"$ | | | M1 | for a complete method<br><br>There are other methods and marks should be awarded for a complete method that should give the correct area |
| | *Working required* | 426 | | A1 | (dep on M1) allow 420 – 427 from correct working |
| | | | | | **Total 5 marks** |

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## Question 23

*Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 10*

| | | | | | |
|---|---|---|---|---|---|
| **10** | $23 \times 4.7$ (= 108.1) oe | | 5 | B1 | (indep)<br>May be embedded in $23 \times (4.7 + 2.5)$ (= 165.6) |
| | $\sin 30 = \frac{(x)}{5}$<br>or<br>$\frac{(x)}{\sin 30} = \frac{5}{\sin 90}$ oe<br>where $x$ = height of trapezium | $5\cos 30 \left(= \frac{5\sqrt{3}}{2} = 4.33...\right)$<br>**and**<br>$(x^2 =) 5^2 - "5\cos 30"^2 (= 6.25)$ | | M1 | |
| | $(x =) 5\sin 30 (= 2.5)$ oe<br>or<br>$(x =) \frac{5}{\sin 90} \times \sin 30 (= 2.5)$ oe | $(x =) \sqrt{5^2 - "5\cos 30"^2} (= 2.5)$ | | M1 | |
| | $\frac{1}{2} \times (11+23) \times "2.5" (= 42.5)$ oe or $\left(\frac{1}{2} \times "2.5" \times (23-11)\right) + (11 \times "2.5") (= 42.5)$ oe or<br>$\left(\frac{1}{2} \times "2.5" \times (23-11-"4.3")\right) + (11 \times "2.5") + \left(\frac{1}{2} \times "2.5" \times "4.3"\right) (= 42.5)$ oe or<br>$(11 \times "2.5") + \left(\frac{1}{2} \times 5 \times (23-11) \times \sin 30\right) (= 42.5)$ oe or<br>$(23 \times "2.5") - \left(\frac{1}{2} \times "2.5" \times (23-11-"4.3")\right) - \left(\frac{1}{2} \times "2.5" \times "4.3"\right) (= 42.5)$ oe or<br>$(23 \times ("2.5"+4.7)) - \left(\frac{1}{2} \times "2.5" \times (23-11-"4.3")\right) - \left(\frac{1}{2} \times "2.5" \times "4.3"\right)$ oe | | | M1 | for a correct method to find the area of the trapezium<br>or<br>the whole shape |
| | *Working required* | 150.6 | | A1 | dep on M1 awrt 150.6<br>Allow 151<br>Accept $\frac{753}{5}$ |
| | | | | | **Total 5 marks** |

## Question 24

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 22*

| | | | | | |
|---|---|---|---|---|---|
| **22** | Ext angle of octagon = 360 ÷ 8 (= 45)<br>**or** Int angle of octagon (8 – 2) × 180 ÷ 8 oe (= 135) | | 6 | M1 | for method to find the size of one exterior **or** one interior angle of a regular octagon |
| | e.g. $10 + 2 \times 10 \times \sin 45$ ($= 10 + 10\sqrt{2}$ or 24.1...)<br>**or** e.g. $\frac{10 \sin 112.5}{\sin 22.5}$ (= 24.1…) | | | M1 | method to find *HE* **or** *AD*<br>22.5 comes from (180 – "135") ÷ 2<br>112.5 comes from "135" – "22.5" |
| | e.g. $10 \times ("10 + 10\sqrt{2}")$ ($= 100 + 100\sqrt{2}$ or 241.4...)<br>**or** 10 × "24.1…" (= 241.4…) | | | M1 | area *ADEH* |
| | e.g. 10 × sin45° (=5 $\sqrt{2}$ or 7.07...)<br>**or** e.g. $\sqrt{10^2 + 10^2 - 2 \times 10 \times 10 \times \cos "135"}$ (= 18.4…)<br>**or** $\frac{10 \sin "135"}{\sin 22.5}$ (= 18.4…) | | | M1 | finds perpendicular height of triangle *ACD* (may be found before, but must realise this is also height of triangle) **or** finds the length of *AC*<br>22.5 comes from (180 – "135") ÷ 2 |
| | e.g. 0.5 × "24.1..." × "7.07…" (= 85.3...)<br>**or** $0.5 \times 10 \times "18.4..." \times \sin 112.5$ (= 85.3…) | | | M1 | finds the area of triangle *ACD*<br>112.5 comes from "135" – "22.5" |
| | | 327 | | A1 | accept 326 – 327 |
| | **Alternative (splitting octagon into triangles and subtracting trapezium and triangle)** | | | | |
| | Ext angle of octagon = 360 ÷ 8 (= 45)<br>**or** Int angle of octagon (8 – 2) × 180 ÷ 8 oe (= 135)<br>**or** one of 8 angles at centre = 360 ÷ 8 (= 45) | | 6 | M1 | for method to find the size of one exterior **or** one interior angle of a regular octagon **or** method to find one angle at centre of octagon when split into 8 equal triangles |
| | e.g. 0.5 × 10 × 5 × tan67.5 (= 60.35...)<br>**or** $0.5 \times \left(\frac{10 \sin 67.5}{\sin 45}\right)^2 \times \sin 45$ (= 60.35...)<br>**or** Octagon = 8 × "60.35" (= 482.8...) | | | M1 | Area of one triangle (one-eighth of octagon) **or** octagon |
| | e.g. $10 + 2 \times 10 \times \sin 45°$ ($= 10 + 10\sqrt{2}$ =24.14..) | | | M1 | Method to find *HE* |
| | $0.5 \times (10 + 10 + 10\sqrt{2}) \times 5\sqrt{2}$ (=120.71...) | | | M1 | Method to find area of trapezium *HEGF* |
| | $0.5 \times 10 \times 10 \times \sin 135°$ (= 35.35...) | | | M1 | Method to find area of triangle *ABC* |
| | | 327 | | A1 | accept 326 – 327 |
| | | | | | **Total 6 marks** |

## Question 25

*Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 25*

| | | | | | |
|---|---|---|---|---|---|
| **25** | [chord $AB$ =] $\sqrt{5^2+5^2-2\times5\times5\times\cos 50}$ **or** $2 \times 5 \times \sin 25$ (= 10sin25 or 4.226...) | | 6 | M1 | oe |
| | [$\angle APB$ =] $\cos^{-1}\left(\frac{4^2+4^2-"4.226..."^2}{2\times4\times4}\right)$ (=63.77...) **or** [$\angle OPA$ =] $\sin^{-1}\left(\frac{0.5\times"4.226..."}{4}\right)$ (= 31.88...) | | | M1 | oe may use other methods but must be a complete method for $\angle APB$ **or** $\angle OPA$ (see below for sine rule) |
| | [Area sector $AOB$ =] $\frac{50}{360}\times\pi\times5^2\left(=\frac{125}{36}\pi \text{ or } 10.9...\right)$ | | | M1 | oe independent |
| | [Area sector $APB$ =] $\frac{"63.77..."}{360}\times\pi\times4^2$ (= 8.90...) | | | M1 | oe NB: 2 × "31.88…" = "63.77…" |
| | $\left(\frac{50}{360}\pi\times5^2-\frac{1}{2}\times5^2\times\sin 50\right)+\left(\frac{"63.77..."}{360}\times\pi\times4^2-\frac{1}{2}\times4^2\times\sin"63.77..."\right)$ | | | M1 | oe (10.9…– 9.57…) + (8.90… – 7.17…) |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | 3.06 | | A1 | allow 3 – 3.1 |
| | ***Alternative version*** (using line of symmetry $OP$ in quadrilateral $OAPB$) | | | | **Total 6 marks** |
| | $[\angle OPA] = \sin^{-1}\left(\frac{5\sin 25}{4}\right)$ (= 31.88..) | | 6 | M1 | oe (see above for cosine rule & trig) |
| | [Area sector $APB$ =] $\frac{2\times"31.88..."}{360}\times\pi\times4^2$ (= 8.90...) | | | M1 | oe |
| | [Area $OAPB$ =] 2 × ½ × 5 × 4 × sin(180 – "31.88.." – 25) (=16.75...) | | | M1 | oe |
| | [Area sector $AOB$ =] $\frac{50}{360}\times\pi\times5^2\left(=\frac{125}{36}\pi = 10.9...\right)$ | | | M1 | oe independent |
| | [Area **R** =] "10.9..." + "8.90..." – "16.75..." | | | M1 | oe |
| | *Working not required, so correct answer scores full marks (unless from obvious incorrect working)* | 3.06 | | A1 | allow 3 – 3.1 |
| | | | | | **Total 6 marks** |

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